

Organizational Systems — Module 01: Systems —

Discussion Questions

Mix of analytical and applied questions for business professionals. Questions 1–10 are analytical; Questions 11–20 are applied.

Analytical Questions

1. How does framing an organization as a "system with a Markov blanket" differ from the traditional view of organizations as hierarchical structures? What does the Active Inference framing reveal that the traditional view obscures?
2. Identify the **sensory states** and **active states** of a university. How do these differ from those of a for-profit technology company? What does this comparison tell you about how organizational boundaries shape behavior?
3. A startup that pivots its business model is, in Active Inference terms, changing its generative model. What internal states must change during a pivot, and which typically resist change? Why?
4. Explain why an organization never has direct access to the "true" state of its market. How does this fundamental uncertainty shape strategic decision-making?
5. Organizational **homeostasis** requires keeping critical variables within bounds. What happens when an organization's homeostatic targets themselves become outdated — for example, when a company still targets pre-pandemic levels of office attendance?
6. In what sense is organizational **culture** an internal state rather than an external state? Can culture ever be a sensory state or an active state? Provide examples.
7. How does the nested structure of organizations (individuals → teams → departments → enterprise) create both advantages and challenges for information processing? Where do communication breakdowns typically occur in this hierarchy?

8. The Markov blanket concept implies that organizations are informationally *partially closed* — they let some signals in and block others. When is this filtering beneficial? When does it become pathological?
 9. Compare Toyota's organizational Markov blanket with that of a typical fast-food franchise. How do differences in boundary permeability explain differences in innovation capacity?
 10. If an organization's sensory states are systematically biased (e.g., customer feedback systems that only capture complaints, not satisfaction), how does this affect the organization's generative model and subsequent decisions?
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Applied Questions

1. Map the Markov blanket of your own organization (or a department within it). List at least 5 sensory states, 5 active states, and 5 internal states. Which boundaries are well-defined? Which are fuzzy?
2. Identify one area where your organization's **boundary is too rigid** (blocking important signals) and one area where it is **too porous** (letting in noise or leaking proprietary information). What changes would you recommend?
3. Choose a recent strategic surprise your organization experienced (a missed forecast, an unexpected competitor move, a regulatory change). Trace how the organization's sensory states did or did not detect the signals beforehand. Was this a failure of sensing, interpretation, or both?
4. Your organization has just acquired a smaller competitor. Using the nested systems framework, describe how the two Markov blankets must merge. Where do you predict friction, and where do you expect synergies?
5. Design a simple "organizational health dashboard" that tracks 4–6 homeostatic variables for your organization. For each variable, define the lower bound, upper bound, and target. How would you detect drift toward dangerous ranges?

6. A consulting firm's primary asset is the knowledge inside the heads of its people. When a senior consultant leaves, what happens to the organization's internal states? How can the firm design systems to preserve institutional knowledge across the Markov blanket?
7. Remote work has fundamentally changed many organizations' Markov blankets. Describe how the shift to distributed work altered the sensory states (what information flows in) and active states (what the organization produces). What new boundaries emerged?
8. You are leading a cross-functional project team. Using the concepts from this module, describe how this temporary team creates its own Markov blanket while remaining nested within the larger organizational system. What are the team's sensory and active states?
9. Many organizations have created "innovation labs" that are deliberately separated from the core business. In Markov blanket terms, what does this separation accomplish? What risks does it create for reintegrating innovations into the main organization?
10. Reflect on a time when your organization's internal model of its environment was significantly wrong. What feedback mechanisms (sensory states) eventually corrected the model? How long did the correction take, and what organizational factors slowed or accelerated it?